

Nuclear Astrophysics

Exercise 1

1. Solar abundances:

- (a) Read the following files: `solar_Lodders2009`, `solar_s.dat`, `solar_r.dat`. These files contain the observed total and inferred *s*- and *r*-process solar abundances respectively. Plot the $\log \epsilon$ solar abundances¹ from the above files vs the atomic number Z . Which elements are created mostly through the *r*-process and which ones mostly through the *s*- process?
- (b) Read the following file: `solar_mfrac_anders_grevesse.txt`. Check if the mass fractions² are normalized correctly, i.e. $\sum_i X_i = 1$. Create three plots of the mass fraction vs atomic number Z , mass number A and neutron number N . What are the two most abundant species in the solar system? What percentage of mass in the solar system do they make up? Locate on the plots the positions of the proton and neutron shell closures (magic numbers).

2. Gaia data:

The Hertzsprung–Russell (H-R) diagram, is a scatter plot of stars showing the relationship between the stars' absolute magnitude M or luminosity L versus their color or effective temperature T .

- (a) Read the Gaia data of 3 million stars within 200 pc from the Sun: `gaia-hrd-dr2-200pc.csv`. The file contains the color BP-RP of the stars and their absolute magnitude M . To import the data you can use:

```
gaiarec = np.recfromcsv("gaia-hrd-dr2-200pc.csv")
bp_rp, mg = gaiarec.bp_rp, gaiarec.mg
```

- (b) Plot the H-R diagram.
- (c) Try to locate on the diagram the Main sequence, the Giant branch and the White dwarfs.
- (d) Add labels to show the stars' luminosity in units of the Sun luminosity. Use the relation $M_* - M_\odot = -2.5 \log_{10}(L_*/L_\odot)$, where $M_\odot = 4.83$. To add a secondary axis to the plot you can use:

```
axx = ax.secondary_yaxis('right', functions=(mag2lum, lum2mag))
```

¹The $\log \epsilon_i$ value of an element i is related to its abundance Y_i via: $\log \epsilon_i = \log_{10} \left(\frac{Y_i}{Y(H)} \right) + 12$, where $Y(H)$ is the hydrogen abundance and the factor 12 is a convention.

²The mass fraction X_i of an element i is related to its abundance Y_i via: $X_i = A_i Y_i$.

3. Binding Energy per Nucleon:

The Atomic Mass Evaluation (AME) project is a long running international effort to provide the most reliable, and up-to-date set of nuclear masses for all known nuclides. It is the standard reference in nuclear physics when people talk about “measured nuclear masses.”

- (a) Read the AME data of the known nuclides from: `AME2020_massround.mas20.dat`. The file header specifies the meaning of each value and symbol.
- (b) Plot the binding energy per nucleon as a function of mass number. For each mass number, only plot the isotopes with the maximum binding energy per nucleon. For isotopes with mass number $A > 35$, plot every other data point to avoid overlap.
- (c) Identify the isotopes with the highest binding energy per baryon and indicate which elements they correspond to.
- (d) Why can elements heavier than iron not be synthesized through fusion?

The Bethe-Weizsäcker (semi-empirical) mass formula is a simple approximation for nuclear masses, based on the liquid drop model: The nucleus is modeled as an incompressible fluid with five contributions:

- *Volume* energy with $a_V = 15.56$ MeV
- *Surface* energy with $a_S = 17.23$ MeV
- *Coulomb* energy with $a_C = 0.69$ MeV
- *Asymmetry* energy with $a_A = 23.28$ MeV
- *Pairing* energy with $a_P = 11.2$ MeV

With this approximation, the binding energy per nucleon of a nucleus with Z protons, N neutrons (and mass number $A = Z + N$) is given as:

$$E_B/A = a_V - \frac{a_S}{A^{1/3}} - a_C \frac{Z^2}{A^{4/3}} - a_A \frac{(N - Z)^2}{A^2} + \frac{\delta(N, Z)}{2A^{3/2}}. \quad (1)$$

The pairing term $\delta(N, Z)$ is a_P for even-even nuclei (even Z and even N), 0 for odd-even nuclei (odd A), and $-a_P$ for odd-odd nuclei (odd N and odd Z).

- (e) Plot the Bethe-Weizsäcker formula on top of the experimental data and check the agreement.